

Revision Questions for Test 3 (Assessment 1) 2026 - Memo

Section A – Multiple Choice

(18 marks)

1. What is displayed on the screen by the following code? (2 marks)

```
DECLARE A, B, C, SUM AS REAL
B = 5
A = 2
C = 2
IF B > A AND A > C AND C > B THEN
    B = A + 2
ELSE
    A = B + 1
END IF
SUM = A + B + C
DISPLAY SUM
```

- a. 9
- b. 8
- c. 13
- d. NONE OF THESE

Answer:

c. 13

2. What is displayed on the screen by the code if **5** and **2** are the input values? (2 marks)

```
DECLARE A, B, SUM AS REAL
ACCEPT A, B
IF (A > B AND A - B = 3) OR (B * 2 = 4) THEN
    SUM = A + B
    DISPLAY SUM
ELSE
    DISPLAY A
ENDIF
```

- a. 5
- b. 7
- c. NONE OF THESE
- d. SUM

Answer:

b. 7

3. What is displayed on the screen by the following code? (2 marks)

```
DECLARE NUM1, NUM2 AS REAL
NUM1 = 20
NUM2 = NUM1 \ 2
IF NUM2 * 5 > 30 AND NUM1 > NUM2 * 3 THEN
    DISPLAY NUM2
ELSE
    DISPLAY NUM1
ENDIF
```

a. 6

b. 4

c. 7

d. NONE OF THESE

Answer:

d. NONE OF THESE

4. What is displayed on the screen by the following code if **18** is the input for AGE and **60** is the input for MARK? (3 marks)

```
DECLARE AGE, MARK AS REAL
DECLARE STATUS AS STRING
ACCEPT AGE, MARK
IF (AGE >= 18 AND MARK >= 50) OR (AGE < 20 AND MARK = 60) THEN
    STATUS = "PASS"
    DISPLAY STATUS
ELSE
    STATUS = "FAIL"
    DISPLAY STATUS
ENDIF
```

a. STATUS

b. FAIL

c. PASS

d. NONE OF THESE

Answer:

c. PASS

5. What is displayed on the screen by the following code if **BMW** is input? (3 marks)

```
DECLARE PRICE AS REAL
DECLARE MSG, CAR AS STRING
ACCEPT CAR
IF NOT CAR <> "BMW" THEN
    PRICE = 550
    MSG = "HIGH"
ELSE
    PRICE = 200
    MSG = "LOW"
END IF
DISPLAY CAR, PRICE, MSG
```

- a. BMW 200 LOW
- b. CAR, PRICE, MSG
- c. NONE OF THESE
- d. BMW 550 HIGH

Answer:

d. BMW 550 HIGH

6. What is displayed on the screen by the following code? (2 marks)

```
DECLARE MIN, MAX AS REAL
MIN = 6
MAX = MIN \ 3
IF MIN - MAX = 4 AND MAX - MIN < 0 THEN
    DISPLAY MAX
    MIN = MAX * 2
ELSE
    DISPLAY MIN
ENDIF
```

- a. 6
- b. 2
- c. MIN
- d. 8

Answer:

b. 2

7. What is the purpose of a flowchart in programming? (1 marks)

- A) To test a solution
- B) To calculate outputs
- C) To store data in tables
- D) To graphically represent program logic

Answer:

D) To graphically represent program logic

8. Which flowchart symbol is used for calculations? (1 marks)

- A) Terminal symbol
- B) Processing symbol
- C) Input/Output symbol
- D) Connector symbol

Answer:

B) Processing symbol

9. In a flowchart, how many lines come out of a decision symbol? (1 marks)

- A) One line out
- B) Three lines out
- C) Two lines out
- D) NONE OF THESE

Answer:

C) Two lines out

10. Which rule must be followed when writing instructions inside flowchart symbols? (1 marks)

- A) Instructions can be placed in any symbol
- B) Instructions must be left outside symbols
- C) Instructions must be placed inside the correct symbol
- D) Instructions are optional

Answer:

C) Instructions must be placed inside the correct symbol

11. What is a trace table mainly used for? (1 marks)

- A) Designing program logic
- B) Drawing flowcharts
- C) Declaring variables
- D) Desk-checking a solution

Answer:

D) Desk-checking a solution

12. Which column is always included in a trace table? (1 marks)

- A) Instruction column
- B) Condition column
- C) Variable column
- D) NONE OF THESE

Answer:

A) Instruction column

Section B - True/False

(10 Marks)

1. Every flowchart must begin and end with a terminal symbol.

True / False

Answer:

True

2. The decision symbol in a flowchart has only one line out.

True / False

Answer:

False

3. All variable names used in a flowchart must be declared in the first processing box.

True / False

Answer:

True

4. Flowlines in a flowchart can leave symbols hanging without connecting back into the chart.

True / False

Answer:

False

5. A trace table is used to test a solution by desk-checking with different sets of input data.

True / False

Answer:

True

6. Each instruction executed in a program must be recorded in the trace table under the "Instruction" column.

True / False

Answer:

True

7. In a trace table, each IF Statement condition in the program must have its own column.

True / False

Answer:

True

8. The output column in a trace table is filled for every instruction, regardless of whether the instruction is not a DISPLAY statement.

True / False

Answer:

False

9. Trace tables are part of the Design stage of the Program Development Cycle.

True / False

Answer:

True

10. Changing variable names when writing a trace table is acceptable as long as the logic remains the same.

True / False

Answer:

False

Section C – Trace Tables

(36 marks)

QUESTION 1 (18 marks)

Draw a trace table for the following pseudocode using **10, 20, and 30** as the input values.

```
DECLARE A, P, Q, T, R AS INTEGER
DECLARE S AS STRING
ACCEPT A, P, Q
IF A <> 10 THEN
    T = 5
    R = 7
    A = A + 1
ELSE
    T = Q - P + P MOD 3
    R = P MOD 5 + Q / A
    S = "A"
END IF
DISPLAY S, A
DISPLAY "R, R"
END
```

Solution:

	Instruction	A	P	Q	T	R	S	A <> 10	Output
1	DECLARE A, P, Q, T, R AS INTEGER	0	0	0	0	0	?	?	
2	DECLARE S AS STRING						Null	?	
3	ACCEPT A, P, Q	10	20	30				?	
4	IF A <> 10 THEN							False	
9	T = Q - P + P MOD 3				8				
10	R = P MOD 5 + Q / A					3			
11	S = "A"						A		
13	DISPLAY S, A								A 10
14	DISPLAY "R, R"								R, R

QUESTION 2 (18 marks)

Draw a trace table for the following pseudocode using the input value: **NETWORKING**.

```
BEGIN
DECLARE RANK AS INTEGER
DECLARE SKILL, HATE AS STRING
DISPLAY "PLEASE ENTER YOUR SKILL:"
ACCEPT SKILL
IF SKILL = "networking" THEN
    RANK = 1
    SKILL = "TESTING"
    HATE = "HATE"
ELSE
    HATE = "LIKE"
    SKILL = "HATE"
END IF
IF NOT RANK > 0 THEN
    SKILL = "PROGRAMMING"
END IF
IF SKILL <> "HATE" OR SKILL = HATE THEN
    HATE = "LOVE"
    RANK = 1
ELSE
    SKILL = "BUSINESS ANALYSIS"
    RANK = 2
END IF
DISPLAY "I", HATE, SKILL, "AND MY RANK IS", RANK
END
```

Solution:

	Instruction	RANK	SKILL	HATE	SKILL = "networking"	NOT RANK > 0	SKILL <> "HATE" OR SKILL = HATE	Output
1	DECLARE RANK AS INTEGER	0	?	?	?	?	?	
2	DECLARE SKILL, HATE AS STRING		Null	Null	?	?	?	
3	DISPLAY "PLEASE ENTER YOUR SKILL:"				?	?	?	PLEASE ENTER YOUR SKILL:
4	ACCEPT SKILL		NETWORKING		?	?	?	
5	IF SKILL = "networking" THEN				False	?	?	
10	HATE = "LIKE"			LIKE		?	?	
11	SKILL = "HATE"		HATE			?	?	
12	IF NOT RANK > 0 THEN					True	?	
13	SKILL = "PROGRAMMING"		PROGRAMMING				?	
15	IF SKILL <> "HATE" OR SKILL = HATE THEN						True	
16	HATE = "LOVE"			LOVE				
17	RANK = 1	1						
22	DISPLAY "I", HATE, SKILL, "AND MY RANK IS", RANK							I LOVE PROGRAMMING AND MY RANK IS 1

Section D – Flowcharts

(36 marks)

QUESTION 1 (18 marks)

The Sunrise Fitness Club wants to apply for sponsorship to run health programs for local residents. To qualify, they need background information about participants. You are asked to design a program that will help them collect and process this information. For each participant, the program should accept the participant's name, age, and membership type.

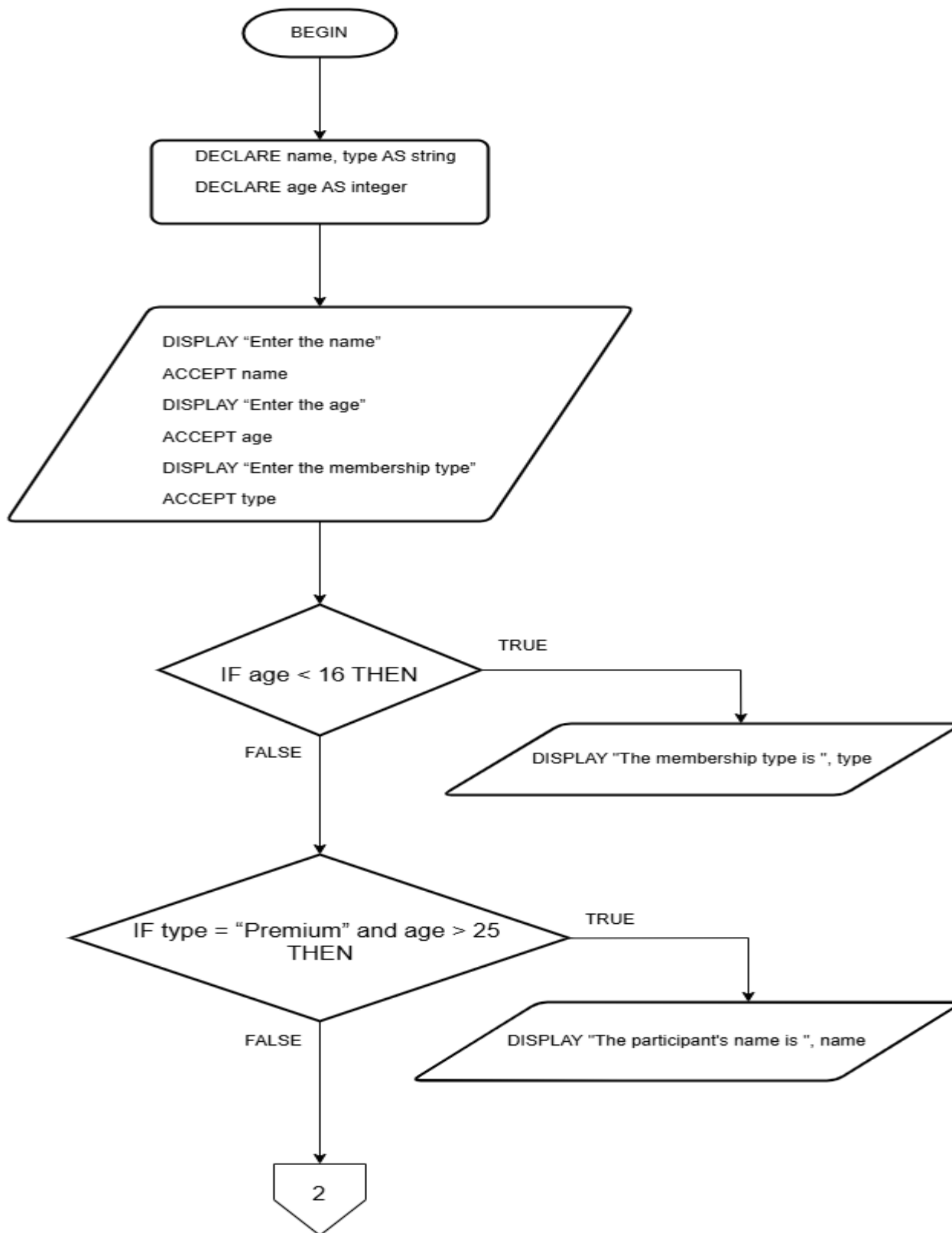
The membership type accepted can only be one of these: **Basic**, **Standard**, or **Premium**.

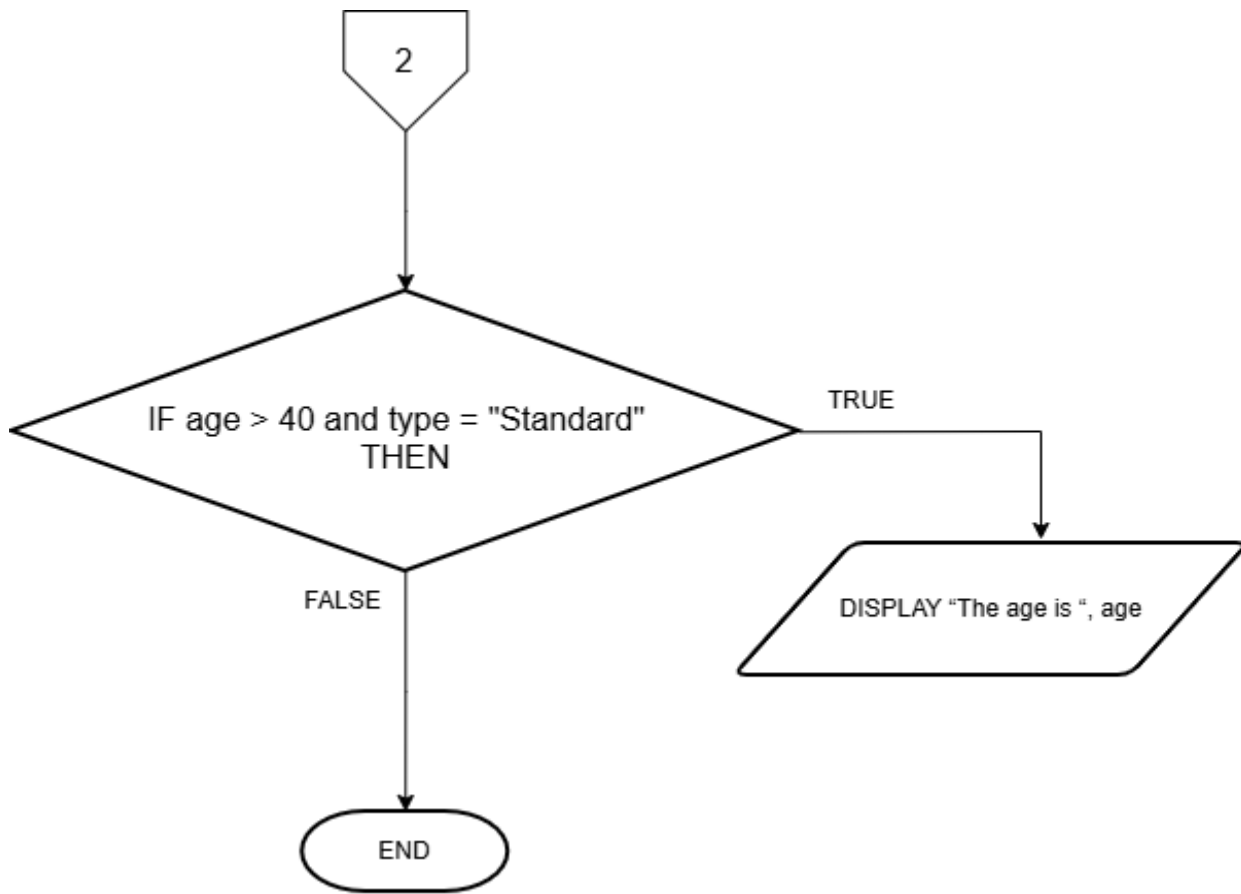
Program Requirements:

- If the participant is under 16 years old, display the participant's membership type.
- If the participant has a Premium membership type and is over 25, display the participant's name.
- If the participant is over 40 and has a Standard membership type, display the participant's age.

Draw a Flowchart for the program.

Solution:





QUESTION 2 (18 marks)

Amathuba Rent a Polo in Mthatha is running a special on Volkswagen Polo cars. The rental price depends on the car type and the number of days rented. The program must get the customer's name, the Polo car type (**Vivo**, **TSI**, **GTI**), and the number of days rented. Calculate and display the rental price based on the table below:

Polo Car Type	Rental Price per Day
Vivo	R250
TSI	R300
GTI	R400

If the customer rents a GTI and the number of days rented is greater than 7, then display the message:
Special Offer: Long-term Polo GTI rental detected.

Draw a Flowchart for the program.

Solution:

